UI/UX audit — Nhill Business, Nhill POS, Nhill Console

> **Date:** 26 October 2026 · **Audited version:** 0.1.0 · **Fixes:** 2.0.0 **Evidence:**
>
> 1. the 50 real screenshots of 0.1.0 (`docs/captures/`, complete journeys against a demo server);
> 2. running the delivered 0.1.0 binaries under Xvfb at 1366×768 and 1024×600 (`docs/assets/audit/v010-*.png`);
> 3. reading the 95 QML files and the `Nhill.Ui` components;
> 4. WCAG 2.1 contrast measurements on the theme colour tokens.
>
> **Limit:** the 2.0.0 fixes are written in the code and checked statically (`scripts/qa/qml-check.py`: 0 errors), but **could not be executed** here (no Qt development files). Their screenshots will be produced by the CI (`scripts/qa/ui-screenshots.sh`, 6 screen sizes). The status of each item says so.

1. Summary

Nhill's visual identity is careful and consistent (night blue, gold, warm paper, IBM Plex, line icons, cards, status chips). The business journeys are complete and understandable. The defects mostly concern **fitting the user's real screen** and **accessibility**:

- the three windows open at a **fixed size** (1280×800, 1280×820, 1320×840), larger than the usable area of the very common 1366×768 laptops, and literally leave the screen on 1024×600;
- **tables** share columns by proportion without a minimum width: in a medium window, dates, numbers and amounts are cut ("26/10/2…", "AV-2026-0…");
- several text and border colours **do not reach the WCAG contrast thresholds**; 11 px and 12.5 px sizes are frequent;
- the **disabled** state of buttons (white text on a 45 % blue) is hard to read and does not explain why the action is unavailable.

2. Method and scale

Each screen was reviewed against the requested list (sizes, resizing, overflow, forms, buttons, truncated text, navigation, consistency, tables, dialogs, scaling, keyboard/focus/contrast, loading/error/empty states, performance).

- **Severity:** Blocking (prevents a task) · High (strong hindrance or risk of error) · Medium (hindrance) · Low (polish).
- **Status:** ✅ fixed in the 2.0.0 code (not executed here) · ⏳ to do · 🔍 to be checked on the CI screenshots.

3. Cross-cutting issues

#	Issue	Evidence	Severity	2.0.0 fix	Status
X1	Fixed-size windows larger than the screen; content cut at the bottom and off-centre on 1024×600; on 1366×768 the window (800 px) exceeds the usable area (≈ 728 px with the taskbar)	`v010-pos-ecran-1024x600.png`, `v010-pos-ecran-1366x768.png`; `PosMain.qml`, `Main.qml`, `ConsoleMain.qml`	**High**	Initial size = min(ideal size, 94 % × 92 % of the screen), centred, opened maximised when the screen is below 1100×700; minimums bounded by the screen size	✅🔍
X2	POS minimum of 1000×640: unusable on the 1024×600 screens of small registers	`PosMain.qml`	High	Minimum 860×560; adaptive selling screen (see P2)	✅🔍
X3	Table columns without a minimum width: cut text with no way to read it	screenshots `business-14`, `business-15`, `console-a05`	High	`DataTable`: per-column minimum width (or 72 px per unit), **horizontal scrolling**, tooltip on any cut cell	✅🔍
X4	Tables not usable with the keyboard	`DataTable.qml` (rows not focusable)	Medium	Tab enters the table, arrows/Page Up/Down move, Enter or Space opens; current row outlined	✅
X5	Insufficient contrast: “faint” text #8A9097 on #F7F5F0 = **2.96:1** (placeholders, help, secondary headings); field borders #CFC9BD on white = **1.65:1** (WCAG 1.4.11 requires 3:1); dark theme “faint” 3.7:1	WCAG computation (`Theme.qml`)	High (accessibility)	`textFaint` #62686F (≥ 4.6:1 on every surface), `inputBorder` #8A8376 (3.45–3.76:1), dark #8392A2 (≥ 5:1) and #5D7389; “success” green 5.27:1	✅
X6	Small font sizes (11 px, 12.5 px) for table headers, help text, captions	`Theme.qml`	Medium	12 px and 13 px	✅🔍
X7	Disabled button: white text on 45 %-opacity blue, hard to read; no hint of the reason	screenshots `business-03`, `business-04`, `pos-p01`	Medium	Explicit disabled colours (`disabledBg` / `disabledText`); error message under an invalid e-mail; Enter in the password field only submits when both fields are filled	✅
X8	Focus ring invisible on primary buttons (#2D6A8F border on #173F5F blue)	`FbButton.qml`	Medium	2 px ring **outside** the button, visible on every variant	✅

#	Issue	Evidence	Severity	2.0.0 fix	Status
X9	Accessible names: fields announce their placeholder (often empty) instead of their label; text areas capture Tab	`FbTextField.qml`, `Field.qml`, `FbTextArea.qml`	Medium	The `Field` label becomes the accessible name; Tab/Shift+Tab leave text areas; icon buttons named by their tooltip	✅
X10	Passwords: no way to check what was typed	every sign-in/creation screen	Medium	Automatic “Show / Hide password” button on every password field	✅
X11	Side menus overflowing in short windows; cut labels (“Paramètres de la plateform…”, “Événements des prestataire…”) without tooltip	screenshots `console-a*`, `Shell.qml`	Medium	Scrollable Business menu; tooltips on cut labels (Business and Console); visible keyboard focus in the console	✅
X12	Form fields whose width depends on the typed text: unbalanced grids (“Legal name” squeezed to “IPLE DÉMO SARL”)	screenshot `business-16`	Medium	Stable preferred width for fields (`Field`): equal columns	✅🔍
X13	Drop-down lists, calendar and notifications that can leave the window	`FbComboBox`, `DateField`, `NotificationsButton`	Medium	Window margins on these popups (automatic repositioning)	✅
X14	Short-lived error messages (4.2 s): a failure message may vanish before it is read	`Notifier.qml`	Medium	Errors and warnings 8 s, at most 4 messages, click to dismiss, “alert” role for screen readers	✅
X15	Information by colour only? No: status chips always carry a text (“Paid”, “Unpaid”)	screenshots	—	Compliant	—
X16	Animations	100–140 ms opacity transitions only	—	Compliant (sober, negligible cost)	—

4. Nhill Business

#	Screen	Issue	Severity	Fix / recommendation	Status
B1	Welcome / sign-in	“Sign in” button disabled without explanation; invalid e-mail shown by a red border only	Medium	Message “Enter a valid e-mail address (for example name@company.cd)” under the field	✅
B2	Creation / joining	10-character password rule	—	8 characters (requirement), help updated, show/hide	✅
B3	Top bar	Search box with a fixed 420 px width	Low	160–420 px depending on the space	✅
B4	Sidebar	Establishment and user names cut (“Quincaillerie Exem…”)	Low	Tooltip with the full names	✅
B5	Payments	“Date” column repeating the payment method (“26/10/2026 · Esp…”) although a “Method” column exists	Medium	Date only; minimum widths (date, document, amount)	✅
B6	Purchases	“Reference” column repeating the date (cut)	Low	Reference only; date in its own column; phone subtitle = date	✅
B7	Reports	11 columns: everything cut	High	Minimum widths per data type (date 104 px, amount 128 px, number 150 px) + horizontal scrolling	✅🔍
B8	Activity log	Raw actions displayed (“action.pos.session_closed”) and technical details (“by_owner : oui · cash_counted : 25656…”, UUIDs)	High (readable traceability)	51 FR/EN labels added to the server (POS, support, console); UUIDs and minor-unit amounts hidden from the summary; “Point of sale” and “Help requests” filters	✅
B9	New invoice	Dense line editor (6 fields + tax on a 2nd row); discreet “Incomplete line” label	Medium	Recommendation: card line editor in narrow windows; not changed (regression risk without the ability to run it)	⏳
B10	Settings	7 tabs in a row: possible overflow in a narrow window	Low	Recommendation: scrollable tabs (`TabStrip`)	⏳
B11	Dashboard	Readable indicators; 11 px captions	Low	Sizes raised (X6)	✅
B12	Empty / loading states	Present and consistent (`EmptyState`, indicators)	—	Compliant	—

5. Nhill POS

#	Screen	Issue	Severity	Fix	Status
P1	Sign-in	Window larger than the screen, form off-centre and cut on 1024×600	High	X1/X2	✅🔍
P2	Sale	Fixed 420 px ticket panel; top bar overflowing below 1200 px	High	Panel 340–480 px (38 % of the width); user name hidden and Help / Sign out buttons as icons (with tooltip) below 1200 px	✅🔍

#	Screen	Issue	Severity	Fix	Status
P3	Ticket	Item names cut (“Ciment 50 kg …”, “Livraison (dé…”)	Medium	Names on two lines, adaptive row height	✅
P4	Ticket	Quantity changed with ‹ › chevrons (ambiguous meaning)	Medium	“−” and “+” buttons, tooltips “Decrease the quantity” / “Remove the line” / “Increase the quantity”, quantity announced to screen readers	✅
P5	Sale	No keyboard shortcut for a cash register	Medium	**F2**: back to the search/scan field; **F9**: pay (when the ticket is not empty); reminder in the field and the tooltip	✅
P6	Session	Header with 5 buttons + title: overflow below ≈ 1180 px; 4 indicators on one row	Medium	Secondary buttons as icons, title reduced and shortened when needed; indicators on 2 columns below 1000 px	✅ 🔍
P7	Session	“Time” column showing date and time	Low	“Date and time” column, minimum width	✅
P8	Ticket (return)	**TypeError** when “Return” fires before the ticket is loaded (reported by the console in 0.1.0)	Medium	Guard; no action until the details are loaded	✅
P9	Closing	Closing without entering the cash: the server refuses (`cash_count_required`) with a generic message	Medium	Check before sending + explicit message “Enter the cash counted in the drawer (0 if the drawer is empty)”	✅
P10	Return dialog	“Qty to r…” placeholder cut	Low	Recommendation: column label above the fields	⏳

6. Nhill Console

#	Screen	Issue	Severity	Fix	Status
C1	All	Fixed 1320×840 window	High	X1	✅ 🔍
C2	Menu	15 entries, cut labels, no focus ring	Medium	X11	✅
C3	Sign-in	Mandatory TOTP code (2FA)	—	Kept: platform-privileged accounts (see security audit); 8-character minimum password	✅
C4	Requests, errors, audit	Cut columns (“Quincaillerie Exemple …”, “qrc:/Nhill/Pos/pages/…”)	Medium	X3 (tooltips, scrolling)	✅ 🔍
C5	Server installation	Clear 4-step wizard; 11 px help text	Low	X6	✅

7. Design system (2.0.0)

Element	Rule
Colours	`Theme.qml` tokens: surfaces (`bg`, `surface`, `surfaceAlt`, `sunken`), text (`text` 16:1, `textMuted` ≥ 5.8:1, `textFaint` ≥ 4.6:1), brand (`primary` #173F5F, `accent` #8E6519), states (`success`, `warning`, `danger`, `info` + *soft* versions), `inputBorder` ≥ 3:1, `disabledBg/Text`, `focus`
Typography	Embedded IBM Plex Sans / Mono; scale 12 / 13 / 14 / 16 / 20 / 26 px; weights 400/500/600/700
Spacing	4 / 8 / 12 / 16 / 24 / 32 px (`s1` … `s6`); radii 8 and 12 px

Element	Rule
Controls	Height 38 px (44 px in compact/touch mode); primary / secondary / ghost / danger buttons; 18–20 px icons
Focus	2 px `focus` ring, outside for buttons, border for fields and table rows
States	Explicit disabled; busy (indicator inside the button); empty (`EmptyState` with an action); error (text under the field + 8 s message)
Tables	Per-column minimum width, horizontal scrolling, tooltips, keyboard, cards on phones
Windows	Size fitted to the screen, centred, minimums ≤ screen, maximised below 1100×700 (1024×680 for Business)
Dialogs	Width ≤ window − 48 px, height ≤ window − 48 px, scrolling content, full screen in phone mode
Language	Every visible string goes through `qsTr` and exists in French (automatic check)

8. What remains (priority)

1. 🔍 Review the screenshots produced by the CI at the 6 screen sizes and fix any gap.
2. ⏳ Card editor for invoice lines in narrow windows (B9).
3. ⏳ Scrollable settings tabs (B10); column labels in the return dialog (P10).
4. ⏳ Test with a screen reader (Orca on Linux, NVDA on Windows) and with 125 % and 150 % system scaling.
5. ⏳ Observation sessions with 3 to 5 shopkeepers (pilot programme): time to issue an invoice, ring up a sale, close a register.